Python Automata Library (PAL) Cheat Sheet

import PythonAutomataLibrary as pal · Safe mode is default. Call **pal.FastMode**() before constructing any PAL object. Put substantive model kernels under **@pal.njit(cache=True)**.

Create state

`pal.NewGrid(dims, dtype)` — Create one value per site; supports bool, fixed-width integer, and float dtypes.

`pal.NewAgentGrid(dims, numAgentProps=0, isStackable=False)` — Create mobile agents with per-agent properties. isStackable permits multiple agents per site; dims=() makes a nonspatial collection.

`pal.NewPopGrid(dims, capacity=None)` — Store integer population counts per site; capacity limits total population.

`pal.NewPDEgrid(dims)` — Create a continuous field with queued synchronous updates and diffusion/advection operators.

`pal.NewIList()` — Create a mutable integer list for collecting/reusing query results.

`pal.NewMultinomial()` — Create a sampler for binomial/multinomial population draws.

Shared lattice geometry

Properties — xDim / yDim / zDim: axis sizes; nDims: number of axes; wrapX / wrapY / wrapZ: wrapping flags.

`ToI(x...)` — Convert coordinates to a linear site index; inverse methods are ItoX/Y/Z(i).

`grid.Hood(hood, x...)` — Apply relative coordinate offsets, e.g. ((x1,y1),(x2,y2),...), at x.... Special-case hoods: **pal.MooreHood**(*), **pal.VonNeumannHood**(*), **pal.CircleHood**(*,rad,*). *...

Grid / common indexing

Indexing — grid[i] reads/writes by linear index; grid[x...] uses dimensional coordinates.

Slices — Return detached NumPy copies, not views; write back explicitly to change the grid.

`Box(x1,x2,...)` — Select a rectangular region: (xLow,xHigh,yLow,yHigh,...), with exclusive upper bounds.

Shared window API

GIF recording — win.StartGif(path,delay=100) starts recording; win.AddGifFrame(...) appends a frame; win.StopGif(...) finishes. Supported by PixWindow and OpenGLWindow; optional frame...

Draw / output

`pal.StartPixWindow(...)` — Create a pixel buffer and display window; headless=True supports noninteractive rendering.

`pix[x,y] = RGB` — Set a pixel color in the shared pixel buffer.

Color utilities — pal.RGB(r,g,b) packs red, green, and blue channels into a color; pal.GetR(color), pal.GetG(color), and pal.GetB(color) extract channels; pal.ColorScale(...) maps values to colors.

PixWindow controls — win.Update() refreshes the display; win.Save(path,block=True) saves an image; win.Close() releases the window.

`pal.StartOpenGLWindow(xDim,yDim,zDim=None,width=800,height=800,...)` — Create the renderer and return (draw, win): draw collects primitives; win handles display, updates, and output. Omit zDim for 2D; set it for 3D. headless=True supports...

`draw.Circle(rad,color,x,y,z=None)` — Add a circle or sphere-like primitive with the given radius and color.

`draw.Box(xLen,color,x,y,z=None,yLen=None,zLen=None)` — Add a box centered at a position; specify axis lengths to control its size.

`draw.BoxSQ(color,x,y,z=None)` — Add a box aligned to a lattice site.

`draw.Line(width,color,x1,y1,x2,y2,z1=None,z2=None)` — Add a line segment between two 2D/3D points.

`draw.Borders(width,color)` — Draw grid borders with the requested line width and color.

`win.Camera(x,y,z,yaw=None,pitch=None)` — Set the camera position and optional yaw/pitch angles for a 3D scene.

OpenGL frame — win.Background(color) sets the background; draw.Clear() clears collected primitives; win.Clear() clears the scene; win.Update() renders.

OpenGL output — win.IsOpen() checks the window; win.Save(path,block=False) saves an image; win.Close() releases resources.

Lists + randomness

IList methods — Append(i) adds an integer; Clear() empties the list; Random() picks an entry; Shuffle() reorders entries; All() copies the list; Iter() traverses without copying.

RNG methods — pal.Seed(seed) seeds PAL's RNG; pal.Random() draws uniform [0,1); pal.RandInt(n) draws an integer in [0,n). PAL's RNG is separate from NumPy's.

Multinomial methods — m.Binomial(n,p) draws a binomial count; m.Setup(...) configures multinomial probabilities; m.Sample(...) draws counts.

AgentGrid

`NewAgentSQ(x...)` / `MoveSQ(a,x...)` — Create/move agents by integer site coordinates; linear-site variants are also available.

`NewAgent(x...)` / `Move(a,x...)` — Create/move agents by continuous coordinates (off-lattice).

`grid[a,p]` — Read or write property p for agent a.

Agent position — I(a) gives the linear site index; XSQ/YSQ/ZSQ(a) give lattice coordinates; X/Y/Z(a) give continuous coordinates.

`Alive(a)` / `Dispose(a)` — Check whether an agent is still valid, or remove it from the grid.

`GetPop()` — Count agents currently in the grid.

`AgentsAt(x...)` / `LastAgent(x...)` — Query all agents at a site or just its last agent; useful for stackable grids.

`AgentsInRadius(...)` — Find nearby agents, respecting periodic boundaries.

`counts[x...]` — Read site occupancy counts without modifying them.

Iteration — agents.All() returns a detached snapshot, safe for iteration while adding or disposing agents.

`DispWrapX...(p1,p2)` — Compute the shortest displacement along a periodic axis.

PopGrid + PDEgrid

Synchronous updates — Add(v,...) queues a change; Update() applies queued changes together; Reset() clears current and pending values. Indexed assignment changes current state immediately.

`pop.GetPop()` — Return total population across all sites.

`pop.All()` — Return occupied site indices (not their counts).

PDE setup — field.SetTimeSpaceStep(dt, dx[,dy,dz]) sets time and axis spacing; explicit updates require stable time steps.

Diffusion methods — Diffusion(...) uses the standard scheme; DiffusionMask(...) applies a mask; DiffusionField(...) uses spatially varying coefficients; DiffusionInterfaces(...) specifies...

Advection methods — Advection(...) uses constant velocity; AdvectionField(...) uses spatially varying velocity; AdvectionInterfaces(...) specifies interface velocities.

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Full details: API Guide · API Reference